The Fractal Universe: A Cross-Scale UQSH Interpretation of Matter, Energy, and Coupling Regimes

Ukshin Q. Rexhepi
Tübingen, Germany
ukshin.rexhepi@gmail.com

Conceptual Origin November 8, 2025
March 19, 2026 on Figshare

https://doi.org/10.6084/m9.figshare.31814005

Abstract

This work develops a cross-scale interpretation of physical reality within the framework of the Universal Quantum Foam Hypothesis (UQSH). The starting point is the assumption that the universe is not a system consisting of discrete fundamental objects, but rather a dynamic field medium whose organization unfolds recursively across many scales.

In this framework, observable structures do not appear as independent objects, but as stable, coupling-capable organizational states of field energy within a respective regime. Particles, atoms, molecules, as well as astrophysical structures such as stars and galaxies are thereby interpreted as different manifestations of the same underlying field dynamics.

A central principle of this work is the distinction between existing energy organization and observable coupling. Not every energetic structure is directly measurable; only those states become visible whose degree of organization permits coupling to the respective observation regime.

On this basis, a fractal scale analogy is developed in which structural parallels emerge between microscopic and cosmic systems. In particular, it is proposed that the number of stably bound substructures of a system is determined by the coupling capability of its central field node.

The work thus formulates a conceptual framework in which matter is understood as an expression of cross-scale field organization. The goal is to develop a consistent interpretation that describes the observed diversity of physical structures as different stability modes of a universal field medium.

Contents

1 Introduction

Modern physics typically describes nature in terms of specialized theories for different scale ranges. While quantum mechanics treats microscopic systems and relativity theory describes gravitational structures on cosmic scales, a consistent framework is still lacking that interprets these levels as expressions of a common structural organization.

The Universal Quantum Foam Hypothesis (UQSH) proposes to understand the universe as a continuous, dynamic field medium whose organization unfolds recursively across many scales. In this picture, observable structures do not emerge as fundamental building blocks, but as stable organizational states of this field.

A central feature of this interpretation is the observer-relativity of physical structures. Whether a system appears as a "particle," "atom," or "galaxy" depends not solely on its existence, but on its ability to couple to the respective observation regime.

From this follows an important distinction: There is no direct equivalence between existing energy organization and actually observable structure. Many states of the field remain below the coupling threshold and are therefore not immediately accessible to the observer.

On this basis, the present work develops a fractal scale interpretation of physical systems. The goal is to elaborate structural analogies between different scale ranges and to show that the diversity of observable objects can be understood as different stability modes of a unified field medium.

In particular, it is examined to what extent the number of stably bound substructures of a system can be interpreted as an expression of its central field organization. This perspective allows atomic and cosmic systems to be considered within a common conceptual framework.

The work is understood as a conceptual program and makes no claim to a complete mathematical formalization. Rather, a structural approach to the interpretation of physical systems is to be formulated, opening new perspectives on the relationship between scales, energy organization, and observability.

2 Fractal Scale Structure of the Field

Within the framework of the Universal Quantum Foam Hypothesis (UQSH), the universe is understood as a continuous field medium whose organization unfolds recursively across many scales. These scales are not to be interpreted as separate levels, but as nested organizational forms of the same underlying field.

2.1 Distinction from Classical Scale and Fractal Approaches

The scale interpretation used within the UQSH framework differs fundamentally from classical concepts of fractal geometry and scaling theory.

In established approaches, scales are typically understood as geometric or statistical self-similarity structures that emerge from the analysis of observable patterns [2, 3]. Scales in this context are descriptive levels used to characterize complex systems.

In contrast, scales within the UQSH framework are not interpreted as external descriptive levels, but as intrinsic states of field organization itself.

A scale regime here corresponds to a physically effective coupling state of the field in which stability, dynamics, and observability are jointly determined. Scales are thus not merely analytical categories, but real organizational forms of the field medium.

In particular, this means:

- Scales do not exist independently of the system, but are part of its dynamics,
- Transitions between scales are physical reorganization processes,
- Observability is bound to the coupling capability within a regime.

The fractal interpretation developed in what follows is therefore not to be understood as classical geometric self-similarity, but as an expression of cross-scale organizational dynamics of a universal field.

2.2 Scale Hierarchy and Nested Organization

The dynamics of the field can be described as a hierarchical structure in which energy organization stabilizes across different scales. Schematically, the following hierarchy emerges:

subregime fluctuations → particle states → atomic structures → molecular systems → macroscopic matter → astrophysical structures

Each of these levels does not represent an independent ontological domain, but a specific stability form of field organization. The transitions between scales do not occur discretely in the sense of separate systems, but continuously through reorganization of field dynamics.

2.3 Coupling Capability and Observability

A central principle of UQSH is the distinction between existing energy organization and observable structure.

Not every energetic configuration of the field is directly measurable. A structure is perceived as a physical object only when it possesses sufficient coupling capability to the respective observation regime.

This can be formulated formally as a condition:

$$\mathcal{W} \geq \mathcal{W}_{\mathrm{crit}},$$

where $\mathcal{W}$ denotes the organizational strength of a field state and $\mathcal{W}_{\mathrm{crit}}$ denotes the minimal coupling threshold of the respective regime.

Structures with $\mathcal{W} < \mathcal{W}_{\mathrm{crit}}$ continue to exist as part of field dynamics, but remain below the observability threshold.

2.4 Bound Substructures and Effective Coupling Number

The stability of a system within the UQSH framework is determined by the ability to permanently bind additional substructures.

What is decisive here is not merely the existence of substructures, but their respective contribution to the overall organization of the system.

We therefore define the effective coupling number of a system as the weighted sum of the contributions of all substructures:

$$N_{\text{eff}} = \sum_i w_i,$$

where w_i describes the contribution of the respective structure to the overall organization and depends on its coupling strength $\mathcal{W}_i$.

This quantity describes the effective stability of a central field node and accounts for both strongly coupled and more weakly bound substructures.

2.5 Central Field Node and Fractal Analogy

A stable physical state can be interpreted within the UQSH framework as a central field node that binds a certain number of substructures through its organizational strength.

This structure is analogous across scales:

- In the atomic regime, the central field node corresponds to the atomic nucleus, while the bound substructures appear as electrons.
- In the astrophysical regime, the central field node corresponds to a galaxy, while bound substructures appear as satellite systems.

The number of these bound structures is determined by the coupling capability of the central node and not by the mere existence of possible subregime structures.

2.6 Discrete Stability Levels

The binding of substructures within the UQSH framework does not occur as mere continuous attachment, but in preferred stability stages of field organization.

The effective coupling number

$$N_{\text{eff}} = \sum_i w_i$$

can therefore grow discontinuously with increasing central organizational strength $\mathcal{W}_{\text{core}}$ as soon as new substructures reach the necessary coupling capability.

Discrete stability levels in this picture thus do not correspond to mere counting of existing units, but to the emergence of new stable contributions to the overall organization of the system.

2.7 Interpretation as a Fractal Universe

The structure described above suggests interpreting the universe as a fractal organizational system.

In this picture:

- Particles,
- Atoms,
- Molecules,
- Stars,
- Galaxies

are not fundamentally different objects, but different manifestations of the same organizational principle on different scales.

The observed diversity of physical structures thus does not arise from a multitude of independent entities, but from different stability modes of a universal field medium.

This perspective forms the basis for the scale analogy between microscopic and cosmic systems developed in what follows.

3 Scale Analogy Between Atomic and Cosmic Systems

Based on the fractal scale structure developed previously, a structural analogy between microscopic and cosmic systems can be formulated. This analogy does not rest on identical physical interactions, but on common organizational principles of the underlying field medium.

3.1 Central Field Node and Bound Substructures

Both atomic and cosmic systems can be described within the UQSH framework as configurations of a central field node with bound substructures.

- In the atomic regime, the atomic nucleus forms the central field node, while electrons appear as bound substructures.

- In the cosmic regime, a galaxy corresponds to the central field node, while dwarf galaxies and satellite systems appear as bound substructures.

In both cases, the number of these substructures is not determined by their mere existence, but by their coupling capability to the central field node.

3.2 Effective Coupling Number and Structural Correspondence

The effective number of bound substructures results from the weighted sum of all contributions, where the coupling strength of individual structures is taken into account:

$$N_{\text{eff}} = \sum_i w_i.$$

This quantity plays the role of an effective electron coupling in the atomic regime, while it corresponds to effective satellite coupling in the cosmic regime.

This yields a structural correspondence:

$$\text{Electrons} \longleftrightarrow \text{Dwarf Galaxies.}$$

3.3 Coupling Strength and Central Organization

The ability of a system to bind substructures is determined by the organizational strength of its central field node.

In the atomic regime, this corresponds to the proton number Z, while in the cosmic regime the central field organization is characterized by the overall structure of the galaxy.

Within the UQSH interpretation, it can therefore be written:

$$Z \;\sim\; N_{\text{eff}}.$$

This relation does not describe exact equality, but a structural correspondence between central organizational strength and number of bound substructures.

3.4 Subregime Structures and Coupling Thresholds

A decisive aspect of this analogy is the existence of substructures below or near the coupling threshold.

All existing structures contribute to the overall stability of the system, but not to the same extent. Strongly coupled structures shape the directly observable main organization, while more weakly bound or subregime states contribute primarily to background stabilization of the field node.

The effective coupling number is therefore not a mere count of visible units, but a measure of the entire weighted field organization:

$$N_{\text{eff}} = \sum_i w_i.$$

This distinction is particularly relevant in the cosmic regime, since many weakly bound or transient structures do not form the dominant satellite population, yet nevertheless contribute to the stability of the overall system.

3.5 Elemental Analogy of Cosmic Systems

Based on these considerations, an analogy between chemical elements and cosmic structures can be formulated.

An element in the atomic regime is characterized by its proton number. Analogously, a cosmic system can be described by its effective coupling number.

The Milky Way can in this sense be interpreted as a system whose effective satellite number corresponds to a certain stability level.

Taking into account the coupling threshold, it follows that not the mere total number of all observed dwarf galaxies is decisive, but their weighted coupling effect within the overall organization of the system.

This perspective allows galaxies to be interpreted as different stability modes within a cross-scale organizational structure, analogous to the classification of chemical elements.

3.6 Interpretation

The scale analogy developed here suggests that the structure of the universe is determined by a unified organizational principle.

Different physical systems do not appear as fundamentally separate categories, but as manifestations of a common fractal pattern of field organization.

This interpretation opens the possibility of investigating structural relationships between seemingly separate areas of physics on a conceptual level.

4 Application: Classification of the Milky Way in the UQSH Framework

The scale analogy developed previously makes it possible to interpret concrete astrophysical systems within the UQSH framework as fractal correspondences of atomic structures. In what follows, this perspective is applied exemplarily to the Milky Way.

4.1 Central Field Organization of the Milky Way

The Milky Way represents a stable field node in the cosmic regime, whose organization is characterized by a combination of mass distribution, rotational structure, and dynamical stability.

Within the UQSH framework, this structure is interpreted as a central vortex state capable of permanently binding a certain number of substructures.

The strength of this central field node is described by an effective organizational quantity $\mathcal{W}_{\text{core}}$.

4.2 Satellite Systems and Effective Coupling Number

The Milky Way possesses a multitude of dwarf galaxies and satellite systems. Within the UQSH framework, however, it is crucial that these structures do not all contribute equally to the effective coupling number.

A distinction is made between:

- strongly coupled satellite systems,
- more weakly bound or transient structures,
- subregime precursors near or below the coupling threshold.

The effective coupling number is therefore understood not as a mere counting quantity, but as a weighted sum of all contributions:

$$N_{\text{eff}} = \sum_i w_i.$$

This quantity describes the total organizationally effective contribution of the satellite population to the stability of the galactic field node.

4.3 Estimation of the Effective Satellite Number

Taking into account the coupling threshold, it can be assumed that all substructures contribute to the overall stability of the system, but to different degrees.

While strongly coupled satellite systems form the dominant, directly observable structure, more weakly bound and subregime structures also contribute to the dynamic stabilization of the field node.

However, these contributions are not equivalent. What is decisive, therefore, is not the mere existence of substructures, but their respective coupling strength and their contribution to the overall organization of the system.

This suggests interpreting the effective coupling number not as a mere counting quantity, but as a weighted sum of the contributions of all substructures:

$$N_{\text{eff}} = \sum_i w_i,$$

where w_i describes the contribution of the respective structure to the overall organization and depends on its coupling strength.

Thus N_{eff} is not a measure of the mere number of visible substructures, but an expression of their total organizationally effective coupling. Even more weakly bound or subregime structures contribute to stability, but with less weight than strongly coupled satellite systems.

While the total number of possible satellite structures can be significantly higher, the effective number reduces to those systems that:

- are bound long-term,
- possess sufficient internal organization,
- remain dynamically stable in the galactic potential.

4.4 Elemental Analogy of the Milky Way

Based on the effective coupling number, the Milky Way can be interpreted as a fractal correspondence of a chemical element.

It holds that:

$$Z_{\mathrm{gal}} \sim N_{\mathrm{eff}}.$$

The Milky Way thus does not simply correspond to a system with maximal substructure number, but to a field node whose character is determined by the weighted coupling of its substructures.

This interpretation is consistent with the assumption that observable main structures and more weakly coupled background structures jointly determine the effective stability of the system.

4.5 Interpretation in the UQSH Framework

The classification presented here shows that the structural diversity of cosmic systems can be understood as an expression of different stability levels of a universal field medium.

The Milky Way appears in this picture as a stable but not maximally saturated field node whose effective coupling number is determined by the dynamics of its satellite systems.

This perspective allows astrophysical systems to be integrated into a unified scale structure and interpreted as part of a fractal organizational principle.

4.6 Outlook

The classification undertaken here represents a first conceptual application of UQSH. A further analysis could investigate to what extent systematic relationships between central field organization and satellite population can be identified across different galaxy types.

In particular, it would be necessary to clarify whether a universal scaling relation between central organizational strength and effective coupling number can be formulated that is valid independent of the scale considered.

5 Universal Scaling Relation and Fractal Stability

The scale analogy developed previously suggests that the structure of physical systems is not random, but follows a general organizational principle that is valid across scales.

5.1 Universal Organizational Relation

Within the UQSH framework, the stability of a system can be described by the relationship between central field organization and effective coupling structure.

Formally, this can be expressed as a scaling-like relation:

$$N_{\text{eff}} = \sum_i w_i \ \sim \ \mathcal{W}_{\text{core}}^{\eta},$$

where $\mathcal{W}_{\text{core}}$ is the organizational strength of the central field node and η is a scale-dependent exponent.

This relation describes that with growing central field organization, the ability to bind and stabilize substructures increases.

5.2 Fractal Stability Across Scales

The same structural relationship appears in different physical regimes:

- In the atomic regime, central organization determines the number of bound electrons.
- In the molecular regime, it determines the stability of chemical bonds.
- In the astrophysical regime, it determines the number and stability of satellite systems.

This recurrence of the same organizational pattern suggests that these are not independent phenomena, but different manifestations of the same underlying field dynamics.

5.3 Saturation and Stability Limits

The binding capability of a system is not unlimited. With increasing organizational strength, saturation effects occur.

This manifests in the fact that additional energy no longer leads primarily to the binding of new substructures, but to the reorganization of existing structures or to the formation of new dynamic forms.

Formally, this means that the effective coupling number does grow, but not arbitrarily linearly with total energy.

5.4 Interpretation in the UQSH Framework

Within the UQSH framework, the relation

$$N_{\text{eff}} = \sum_i w_i$$

describes not a pure counting quantity, but a measure of the total organizationally effective field structure.

The observable diversity of physical systems thus results from different realizations of this relation on different scales.

The fractal nature of the universe therefore does not consist in geometric self-similarity in the classical sense, but in the recurrence of the same organizational principle under changed energetic and dynamic conditions.

5.5 Significance for Physical Interpretation

This perspective shifts the focus from individual objects to the underlying organizational dynamics of the field.

Particles, atoms, and galaxies appear in this picture not as fundamentally different entities, but as different stability states of a universal field medium.

The classification of physical systems therefore occurs not primarily via their size or composition, but via their effective coupling structure and its stability.

6 Scale Analogy Between Atomic Bonds and Galactic Structures

The fractal scale structure developed in the previous section can be further concretized by directly relating atomic systems and galactic structures.

Within the UQSH framework, both are not understood as different physical categories, but as manifestations of the same underlying field organization on different scales.

In what follows, this analogy is systematically developed.

6.1 Elements as Energy Levels of Field Organization

In the atomic regime, chemical elements are characterized by their proton number and their total energy.

Within the UQSH framework, this corresponds to different stability levels of field organization.

An analogous structure appears in the cosmic regime:

Atomic Scale	**Cosmic Scale**
Light elements (H, He)	Low total energy
Medium elements (C, N, O)	Medium organizational energy
Heavy elements (Fe, Ni)	High organizational energy
Very heavy elements (U, Pu)	Extreme energy (unstable)

This correspondence suggests that chemical elements can be interpreted as discrete stability levels of a universal field.

6.2 Central and Peripheral Structures

The atomic structure can be directly mapped onto galactic systems within the UQSH framework:

Atomic Component	**Galactic Analog**	**UQSH Interpretation**
Atomic nucleus	Galactic core	Highly compressed field region
Protons	Central mass	Stable high-energy vortices
Neutrons	Dark matter in core	Neutral stabilization
Electrons	Dwarf galaxies	More weakly bound vortices
Electron shells	Satellite orbits	Discrete energy levels
Chemical bond	Gravitational bridges	Field coupling

This assignment does not describe direct physical equality of forces, but a structural correspondence of organizational forms.

6.3 Coupling Structure and Central Organization

A central result of the scale analogy is the relationship between central field organization and the number of bound substructures.

In the atomic regime:

$$Z \sim N_{\text{eff}}$$

In the cosmic regime, a corresponding structure emerges:

$$\text{Central Core} \sim N_{\text{eff}} = \sum_i w_i$$

This means:

- The stronger the central field organization, the more substructures can be stably bound.
- The effective coupling number is not identical to the total number of existing structures, but a measure of their weighted stability.

This relation represents a cross-scale organizational rule.

6.4 Exemplary One-to-One Correspondences

The scale analogy can be illustrated exemplarily with simple systems.

Hydrogen Atomic scale:

$$1 \text{ proton} + 1 \text{ electron}$$

Simplest stable structure.
Cosmic scale:

$$\text{Galaxy core} + 1 \text{ satellite}$$

Simplest stable galactic configuration.
Interpretation:
Minimal stable field organization with a single coupling.

Helium Atomic scale:

$$2p + 2n + 2e$$

High binding energy and stability.
Cosmic scale:

$$\text{Compressed core} + several stable satellites$$

Interpretation:
Symmetric field configuration with high stability.

Iron Atomic scale:

$$Z = 26$$

Maximum stability per nucleon.
Cosmic scale:

$$\text{Massive core} + many stable satellites$$

Interpretation:
Optimally organized field structure with maximum stability.

Uranium Atomic scale:
Very high proton number, unstable.
Cosmic scale:
Extremely massive systems with unstable dynamics.
Interpretation:
Supercritical field organization with tendency toward reorganization.

6.5 Atomic Bonds and Galactic Coupling

Chemical bonds can be interpreted within the UQSH framework as coupling of multiple field nodes.

- Atoms couple via electron orbitals,
- Galaxies couple via gravitational and dynamic interactions.

Both processes correspond to a stable connection of field organizations.
This suggests:

$$\text{Chemical Bond} \longleftrightarrow \text{Galactic Coupling}$$

Complex molecules correspond in this picture to complex multi-body systems in the cosmic regime.

6.6 Chemical Reactions as Galactic Reorganization Processes

Within the UQSH framework, chemical reactions do not represent isolated microscopic processes, but are expressions of fundamental reorganization of field structures.

An analogous dynamics appears in the cosmic regime during interactions and collisions of galaxies.

Both processes can be interpreted as transitions between different stable organizational states of the same underlying field medium.

6.7 Structural Correspondence

Chemical Reaction	**Galactic Dynamics**
Atoms react	Galaxies interact
Bonds break	Structures are disrupted
New bonds form	New systems form
Energy released/absorbed	Energy redistributed
Transition states	Dynamic instabilities

6.8 Dynamics of Reorganization

In chemical reactions, existing bonding structures are dissolved and new more stable configurations are formed.

Analogously, galactic interactions lead to:

- Distortion of existing structures,
- Redistribution of matter and energy,
- Formation of new stable configurations.

In both cases, this is not the creation of new substance, but a reorganization of existing field structure.

6.9 Energy Flow and Stabilization

Chemical reactions are characterized by energy differences between initial and final states.

Within the UQSH framework, this corresponds to a movement of the system toward more stable field organization.

Analogously, galactic collisions show:

- Energy is not accumulated locally,
- but distributed across different scales,
- often in the form of radiation, dynamics, or structural reorganization.

The energy does not disappear, but transitions into other organizational forms.

6.10 Types of Reorganization Processes

Different chemical reactions can be interpreted analogously to different galactic dynamics:

- Exothermic reactions $\leftrightarrow$ energy-releasing collisions (e.g., radiation outbursts)
- Endothermic reactions $\leftrightarrow$ energy-absorbing restructurings
- Equilibrium reactions $\leftrightarrow$ stable, oscillating systems

These processes describe different paths by which field organization transitions between states.

6.11 Transition States and Instability

Chemical reactions often pass through unstable transition states before reaching a stable final state.

In the cosmic regime, this corresponds to phases of:

- strong structural distortion,
- heightened dynamics,
- temporary instability.

These states are not permanent structures, but necessary intermediate phases of reorganization.

6.12 Interpretation in the UQSH Framework

Within the UQSH framework, chemical reactions and galactic collisions appear as different manifestations of the same fundamental process:

Reorganization of field structure between stable states

The dynamics of physical systems is thus not determined by static objects, but by continuous transitions between organizational forms.

This supports the interpretation that stability is always relative and scale-dependent.

6.13 Periodic Table and Galactic Classification

The periodic table of elements represents an ordered structure in the atomic regime in which elements are classified according to their proton number and energetic properties.

Within the UQSH framework, an analogous classification for cosmic systems can be formulated, based on effective coupling structure and central field organization.

Atomic Classification	**Cosmic Correspondence**
Period (energy level)	Organizational level of galaxy
Group (bonding behavior)	Dynamic coupling structure
Proton number Z	Central field strength $\mathcal{W}_{\text{core}}$
Electron configuration	Satellite structure

This analogy suggests that galaxies can also be classified in an extended "cosmic periodic table," based on their effective coupling number

$$N_{\text{eff}} = \sum_i w_i.$$

Different galaxy types thereby correspond to different stability classes of field organization.

6.14 Reaction Rates and Cosmic Timescales

Chemical reactions proceed with characteristic rates determined by energy barriers and coupling strengths.

Within the UQSH framework, this corresponds to the speed at which field structures change their organization.

Analogously, cosmic processes also possess characteristic timescales:

- Galaxy collisions occur over millions to billions of years,
- Star formation and decay also follow typical timescales,
- Large-scale structure formation occurs over cosmological timescales.

These timescales correspond within the UQSH framework to different reorganization rates of the field.

Formally, this can be written as scale-dependent dynamics:

$$\tau \sim \frac{L}{c_{\text{eff}}},$$

where L is a characteristic scale length and c_{eff} is an effective reorganization speed.

This means:

- Small scales reorganize quickly,
- Large scales reorganize slowly,
- The underlying dynamics remains structurally identical.

Thus chemical reactions and cosmic evolution appear as temporally scaled variants of the same process.

6.15 Binding Energy and Energetic Field Structure

In atomic physics, binding energy describes the stability of a system.

The higher the binding energy, the more stable the structure.

Within the UQSH framework, this quantity can be interpreted as a measure of local field organization.

Analogously, in the cosmic regime:

- Gravitational binding determines the stability of galaxies,
- Dynamic energy determines the structure of satellite systems,
- Energetic coupling stabilizes large-scale structures.

This suggests a structural correspondence:

$$E_{\text{binding}} \longleftrightarrow E_{\text{field organization}}$$

Where:

- High binding energy $\rightarrow$ stable, compact systems
- Low binding energy $\rightarrow$ diffuse, unstable systems

In the extended UQSH picture, energy is not to be understood as an isolated quantity, but as an expression of the strength of field organization.

This connects atomic stability and gravitational structure as two manifestations of the same principle.

6.16 Scale Invariance of Organizational Dynamics

The analogies described previously show that fundamental physical processes appear invariant across scales.

Both:

- atomic bonds,
- chemical reactions,
- and galactic dynamics

follow the same structural principles:

- Organization rather than accumulation,
- Stability through coupling,
- Reorganization as the dominant process.

This suggests that physical reality need not be described by a multitude of independent laws, but by a unified organizational principle that manifests across all scales.

6.17 Scale-Invariant Field Solutions: From Electron to Universe

Within the UQSH framework, it is assumed that the underlying field structure is scale-invariant.

This means that stable organizational states of the field are not bound to a specific physical scale, but can appear as solutions of the same fundamental dynamics on different scales.

A simple example of such a stable state is the electron, which appears in the atomic regime as a compact, locally bound field organization.

In the cosmic regime, stable, locally organized structures also occur, for instance in the form of galaxies or large-scale field nodes.

Within the UQSH framework, this can be interpreted as an indication that both structures represent solutions of the same underlying field equation, but realized on different scales.

Formally, this idea can be expressed as a scale relation:

$$\mathcal{S}_{\text{micro}} \sim \lambda \, \mathcal{S}_{\text{macro}},$$

where λ describes a scale mapping between different organizational levels.

This does not mean that an electron directly is a galaxy, but that both can be described by the same structural class of field solutions.

6.18 Interpretation

The electron appears in this picture as the minimal stable coupling unit of a field node.

The universe itself can analogously be interpreted as the maximal, globally coupled field state.

Between these two extremes exists a continuous hierarchy of stable solutions:

$$\textit{Electron} \rightarrow \textit{Atom} \rightarrow \textit{Star} \rightarrow \textit{Galaxy} \rightarrow \textit{Universe}$$

All these structures differ not in their fundamental nature, but in their scale and effective coupling structure.

6.19 Consequence

This perspective suggests that physical reality can be described by a scale-invariant class of field solutions.

The electron thereby represents the smallest stably observable unit, while the universe can be interpreted as the global solution of the same organizational principle.

The diversity of physical systems thus results from different scalings and coupling states of the same underlying field dynamics.

6.20 Mathematical Indication of Scale-Invariant Field Structure

Within the UQSH framework, it is assumed that the underlying field dynamics is characterized by a scale-invariant structure.

Without specifying a complete field equation, this assumption can be heuristically represented by a general nonlinear evolution equation [9]:

$$\partial_t \phi + \mathcal{N}(\phi) = \mathcal{D}(\phi),$$

where ϕ describes the state of the field, $\mathcal{N}(\phi)$ represents nonlinear interactions, and $\mathcal{D}(\phi)$ dissipative or regulating processes.

6.21 Scale Invariance

A central property of this dynamics is its assumed invariance under scale transformations of the form:

$$x \rightarrow \lambda x, \quad t \rightarrow \lambda^{\alpha} t, \quad \phi \rightarrow \lambda^{\beta} \phi.$$

Under suitable choice of exponents α and β, the structural form of the dynamics is preserved.

This implies that solutions of the equation can exhibit similar organizational patterns on different scales.

6.22 Localized Stable Solutions

Stable physical structures can be interpreted within the UQSH framework as localized solutions of this dynamics.

Formally, this can be conceived as:

$$\phi(x,t) \approx \phi_{\text{loc}}(x - vt)$$

where ϕ_{loc} describes a spatially concentrated, temporally stable configuration.

Such solutions correspond to:

- in the microscopic regime: particle states,
- in the macroscopic regime: astrophysical structures.

6.23 Scaled Solution Families

If $\phi(x,t)$ represents a solution, then under scale invariance there also exists a family of scaled solutions:

$$\phi_\lambda(x,t) = \lambda^\beta \phi(\lambda x, \lambda^\alpha t).$$

This scaled solution describes the same structural state on a different scale.

6.24 Interpretation

The existence of such scale-invariant solution families suggests that:

- Particles, atoms, and galaxies can be interpreted as different scalings of the same field solution,
- Stability is determined by the structure of the solution and not by its absolute size,
- The observed diversity of physical systems emerges from a family of related solutions.

6.25 Relation to Effective Coupling Structure

The effective coupling number

$$N_{\text{eff}} = \sum_i w_i$$

can be interpreted in this context as a measure of the internal structure of such a solution.

It describes how many stably coupled substructures exist within a given field configuration.

6.26 Summary

The present mathematical sketch does not represent a complete theory, but a structural indication.

However, it shows that the assumption of scale-invariant field solutions is consistent with the idea that physical systems of different scales can be described by the same underlying dynamics.

7 Consequences and Testable Predictions

The fractal scale interpretation developed within the UQSH framework leads to a series of consequences for the physical description of systems across different scales.

7.1 Limited Local Concentration

A central result is that local energy organization cannot be increased without limit.

With increasing organizational density, a transition occurs in which additional energy no longer leads to further local compression, but transitions into other organizational forms.

This implies that extreme states do not appear as real singularities, but as transitions into other scale regimes of field dynamics.

7.2 Dominance of Reorganization over Accumulation

The dynamics of physical systems within the UQSH framework is determined not primarily by accumulation, but by reorganization.

This means:

- Energy is not stored locally without limit,
- but distributed across scales,
- where stable structures appear as intermediate states of this reorganization.

This perspective is consistent with observations of turbulent systems as well as astrophysical processes in which energy is transported across many scales [7, 8].

7.3 Cross-Scale Stability Relations

The relation

$$N_{\text{eff}} = \sum_i w_i$$

implies that the stability of a system is not determined by individual dominant components, but by the entire coupling structure of all participating subsystems.

From this follows:

- Systems with similar effective coupling structure show comparable stability properties,
- independent of their absolute size or energetic scale.

7.4 Prediction for Cosmic Systems

For astrophysical systems, it follows that the observable satellite population does not reflect the complete structure.

It is expected that:

- numerous weakly coupled or subregime structures exist,

- which are not directly observable,
- but contribute to the dynamic stability of the overall system.

This could manifest, for example, in unexpected stability properties of galaxies or in deviations from purely mass-based models.

7.5 Prediction for Microscopic Systems

In the microscopic regime, it follows that not all possible energetic states appear as stable particles.

Instead, only those states become observable whose degree of organization permits sufficient coupling to the respective regime.

This suggests that:

- additional subregime states could exist,
- which are accessible experimentally only indirectly.

7.6 Interpretation of Physical Objects

UQSH leads to a fundamental shift in the interpretation of physical objects.

Particles, atoms, and galaxies are not understood as fixed units, but as stable organizational states within a continuous field dynamics.

The observed reality thus results from the selection of those states that can couple to the respective observation regime.

7.7 Summary

UQSH implies that physical systems are determined by universal organizational principles that manifest in the form of cross-scale stability relations.

These principles lead to clear consequences:

- no unlimited local energy concentration,
- dominance of reorganization over accumulation,
- weighted coupling structure as stability measure,
- existence of not directly observable subregime structures.

These statements are in principle testable and form the basis for further empirical investigation of the UQSH framework.

The structural analogies developed previously suggest that UQSH can be formulated not only interpretatively but also dynamically. In what follows, a first effective equation approach is therefore sketched that mathematically indicates the principles of saturating and cross-scale reorganization.

8 A First Concrete UQSH Equation Approach

The principles formulated within the UQSH framework can be expressed in a first step by an effective field equation that encodes three central properties:

- no unlimited local inward concentration,
- cross-scale reorganization of field structure,
- causally limited reorganization speed.

As the simplest effective approach, a scalar order field $\phi(x,t)$ is introduced that describes the local organizational state of the field medium.

A possible UQSH equation then reads:

$$\partial_t \phi + \mathbf{v}[\phi] \cdot \nabla\phi = \nu\Delta\phi + \gamma\,\nabla\cdot\left[\sigma\!\left(\frac{|\nabla\phi|}{S_H}\right)\frac{\nabla\phi}{\sqrt{\varepsilon^2+|\nabla\phi|^2}}\right] + \kappa\,\mathcal{R}[\phi] - \mu\,\phi. \quad (1)$$

Here:

- $\phi(x,t)$: local organizational state of the field,
- $\mathbf{v}[\phi]$: effective advective self-coupling,
- $\nu\Delta\phi$: local smoothing or scale-local dissipation,
- S_H: saturation limit of local gradient organization,
- $\sigma(z)$: activation function of reorganization,
- γ: strength of saturating reorganization,
- $\mathcal{R}[\phi]$: nonlocal reorganization term,

- κ: strength of cross-scale coupling,
- μ: relaxation or decay parameter,
- $\varepsilon > 0$: regularization parameter.

8.1 Activation of Reorganization

The function $\sigma(z)$ is meant to express that reorganization becomes relevant only when local field organization approaches the saturation limit. A simple choice is:

$$\sigma(z) = \frac{z^m}{1 + z^m}, \qquad m \geq 1. \tag{2}$$

For small gradients $z \ll 1$, $\sigma(z) \approx 0$, while for $z \gg 1$ the reorganization dynamics is activated.

8.2 Nonlocal Reorganization Term

The cross-scale dynamics is modeled by a nonlocal term. A natural approach is:

$$\mathcal{R}[\phi](x,t) = \int_{\mathbb{R}^3} K_\ell(x - y)\big(\phi(y,t) - \phi(x,t)\big)\, dy, \tag{3}$$

where K_ℓ is a normalized kernel with characteristic coupling scale ℓ.

This term describes that local over-organization does not remain isolated, but incorporates neighboring regimes into the dynamics.

8.3 Causal Limitation

UQSH requires that reorganization cannot occur arbitrarily fast. This is expressed by demanding that the effective reorganization speed remains bounded from above by c. Formally, this means that the fluxes induced by (1) must satisfy a bound of the form

$$|\mathbf{J}_{\mathrm{reorg}}| \leq c\, \rho_\phi \tag{4}$$

where ρ_ϕ denotes an effective field energy density.

8.4 Interpretation

Equation (1) thus contains four central components:

1. an advective self-coupling,
2. local diffusion or smoothing,
3. a saturating reorganization term,
4. a nonlocal cross-scale coupling.

Within the UQSH framework, this equation describes not mere material motion, but the dynamics of a field medium that responds to local over-organization not with unlimited concentration, but with reorganization.

8.5 Significance for Stable Solutions

Stable particle, atom, or galaxy structures appear in this picture as localized, long-lived solutions of equation (1). Different physical systems thus do not correspond to different ontological categories, but to different stable solution regimes of the same underlying dynamics.

8.6 UQSH Extension of the Navier–Stokes Equation

Within the framework of the previously introduced UQSH field dynamics, a modified form of the Navier–Stokes equation can be formulated that explicitly accounts for the principles of saturating and cross-scale reorganization.

The classical incompressible Navier–Stokes equation reads:

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0. \tag{5}$$

Within the UQSH framework, this dynamics is supplemented by an additional reorganization term:

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + \gamma \, \nabla \cdot \left[\sigma\left(\frac{|\nabla u|}{S_H} \right) \frac{\nabla u}{\sqrt{\varepsilon^2 + |\nabla u|^2}} \right] + \kappa \, \mathcal{R}[u]. \tag{6}$$

8.7 Significance of UQSH Terms

The additional contributions have the following interpretation:

- The saturating term

  $$\gamma \nabla \cdot \left[\sigma\left(\frac{|\nabla u|}{S_H} \right) \frac{\nabla u}{\sqrt{\varepsilon^2 + |\nabla u|^2}} \right]$$

  activates when the local gradient strength reaches a critical threshold S_H. It prevents unlimited inward concentration and leads to redistribution of energy.

- The nonlocal term

  $$\kappa \, \mathcal{R}[u]$$

  describes cross-scale coupling and models the reorganization of dynamics across neighboring scales.

8.8 Nonlocal Coupling

The operator $\mathcal{R}[u]$ can be formulated analogously to the scalar case as a convolution:

$$\mathcal{R}[u](x,t) = \int_{\mathbb{R}^3} K_\ell(x-y)\big(u(y,t) - u(x,t)\big)\, dy, \tag{7}$$

where K_ℓ is a smooth, normalized kernel with characteristic scale length ℓ.

8.9 Physical Interpretation

Equation (6) describes an extension of classical fluid dynamics in which local over-organization does not lead to unbounded gradient intensification, but transitions into cross-scale reorganization through saturating and nonlocal mechanisms.

In contrast to classical Navier–Stokes dynamics, in which possible blow-up scenarios are characterized by uncontrolled gradient development, the UQSH term implements intrinsic stabilization through field structure.

Thus a possible blow-up no longer appears as a physical end state, but as an indicator that the classical equation without reorganization mechanism is incomplete.

8.10 Relation to the Regularity Question

The extension introduced in this work provides a concrete mathematical formulation of the previously heuristically argued UQSH principles.

In particular, the saturating term corresponds to the assumption that local energy organization cannot exceed a maximum density, while the nonlocal term models cross-scale energy distribution.

This translates the idea of limited high-frequency accumulation formulated in the preceding Navier–Stokes approach into explicit dynamics.

9 Conclusion

In this work, a cross-scale interpretation of physical systems has been developed within the framework of the Universal Quantum Foam Hypothesis (UQSH).

Starting from the assumption of a continuous field medium, it has been shown that observable structures need not be understood as fundamental objects, but as stable organizational states of field dynamics within a respective coupling regime.

A central result is the introduction of the effective coupling number

$$N_{\text{eff}} = \sum_i w_i,$$

which describes the stability of a system as the result of the entire weighted coupling structure of its subsystems.

On this basis, a fractal scale analogy was formulated that makes structural parallels between microscopic and cosmic systems visible. Particles, atoms, and galaxies thereby appear as different manifestations of the same organizational principle.

The analysis of the Milky Way showed exemplarily that not the mere number of existing substructures is decisive, but their respective coupling strength and their contribution to the overall organization of the system.

Furthermore, a universal scaling relation was proposed that describes the relationship between central field organization and effective coupling structure.

The consequences derived from this suggest that physical systems are characterized by the following principles:

- limited local energy concentration,
- dominance of reorganization over accumulation,
- stability as the result of weighted coupling structures,
- existence of subregime, not directly observable states.

The present work is explicitly understood as a conceptual framework and makes no claim to complete mathematical formalization.

The goal is to formulate a consistent perspective that interprets the diversity of physical structures as an expression of a unified, cross-scale organizational principle.

Further investigation could focus in particular on the quantitative determination of the weighting factors w_i as well as on the empirical verification of the proposed scaling relation.

UQSH thus opens the possibility of considering seemingly separate areas of physics within a common conceptual framework and identifying new relationships between scales, energy organization, and observability.

It should be emphasized that this extension is not understood as a rigorous solution to the classical Navier–Stokes problem, but as a physically motivated approach that accounts for additional structural dynamics not contained in the classical formulation.

Parts of the editorial work were carried out with the support of AI-based tools.

Citation

Rexhepi, U. Q. (2026). *The Fractal Universe: A Cross-Scale UQSH Interpretation of Matter, Energy, and Coupling Regimes.* Ukshin Q. Rexhepi. Independent Researcher. Tübingen, Germany. `https://orcid.org/0009-0003-4145-5431`
Figshare: `https://doi.org/10.6084/m9.figshare.31814005`

References

[1] Mandelbrot, B. B. (1967). *How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension*. Science, 156(3775), 636–638.

[2] Mandelbrot, B. B. (1983). *The Fractal Geometry of Nature*. Henry Holt and Company.

[3] Barenblatt, G. I. (1996). *Scaling, Self-Similarity, and Intermediate Asymptotics*. Cambridge University Press.

[4] Sornette, D. (2006). *Critical Phenomena in Natural Sciences: Chaos, Fractals, Selforganization and Disorder: Concepts and Tools*. Springer.

[5] Nottale, L. (1993). *Fractal Space-Time and Microphysics: Towards a Theory of Scale Relativity*. World Scientific.

[6] West, G. B. (2017). *Scale: The Universal Laws of Growth, Innovation, Sustainability, and the Pace of Life in Organisms, Cities, Economies, and Companies*. Penguin Press.

[7] Kolmogorov, A. N. (1941). The Local Structure of Turbulence in Incompressible Viscous Fluid for Very Large Reynolds Numbers. *Doklady Akademii Nauk SSSR*, 30, 301–305.

[8] Onsager, L. (1949). *Statistical Hydrodynamics*. Nuovo Cimento, 6(Suppl. 2), 279–287.

[9] Bahouri, H., Chemin, J.-Y., Danchin, R. (2011). *Fourier Analysis and Nonlinear Partial Differential Equations*. Springer.

[10] Rexhepi, U. Q. (2026). *Foundations of the Universal Quantum Foam Hypothesis (UQSH)*. Figshare: `https://doi.org/10.6084/m9.figshare.31539715`